

Tuesday 21st May, 2019



General Instructions

- Reading time – 5 minutes.
- Start each question on a new page.
- Write using black or blue pen.
- NESA approved calculators may be used.
- Show all relevant mathematical reasoning and/or calculations.
- Total possible mark: **50**
- Time allowed: **80 minutes**

Question 1 (8 marks)

- (a) (i) Expand the expression $\left(x - \frac{1}{x}\right)^3$. 1
- (ii) Given that $x - \frac{1}{x} = 5$, use your answer in part (i) to find the value of $x^3 - \frac{1}{x^3}$ without finding the value of x . 1
- (b) Classify the relation $x = \sqrt{9 - y^2}$ as one-to-one, one-to-many, many-to-one or many-to-many. 1
- (c) Make y the subject of the equation $x = y^2 - 6y$. 1
- (d) Consider the functions $f(x) = x^2 - 3x + 2$ and $g(x) = \sqrt{x}$.
- (i) Find $f(g(x))$. 1
- (ii) Solve $f(g(x)) \geq 0$. 3

Question 2 (8 marks)

- (a) Suppose $x^3 - 2x^2 + a \equiv (x + 2)Q(x) + 3$ where $Q(x)$ is a polynomial. Find the value of a . 1
- (b) Solve the inequality $\frac{5}{x-4} \geq x$. 3
- (c) If $g(x) = x^2 + 1$, sketch $y = \frac{1}{g(x)}$ and hence state its range. 2
- (d) Find the Cartesian equation of the curve defined by the parametric equations $x = 2 \tan \theta$ and $y = 3 \sec \theta$. 2

Question 3 (8 marks)

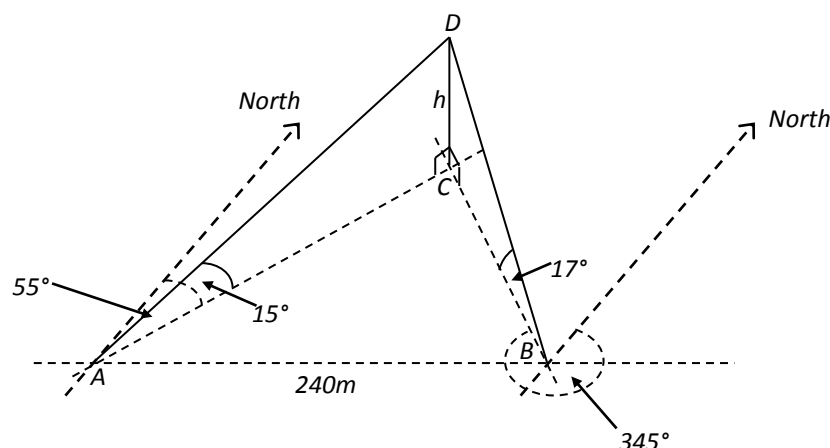
(a) Solve $2\sin x \cos x = \sin x$ for $0^\circ \leq x \leq 360^\circ$.

3

- (b) Ellie walks along a straight road AB . At point A she notices a tower. From this point, the tower is on a bearing of 055° . The top of the tower has an angle of elevation of 15° .

After walking 240 metres, she arrives at point B . From here, the tower is on a bearing of 345° with an angle of elevation to the top of the tower of 17° .

Let the height of the tower be h .



- (i) With the aid of a diagram, show that $\angle ACB = 70^\circ$. 1
- (ii) Find expressions for AC and BC in terms of h . 2
- (iii) Calculate the height, h , of the tower to the nearest metre. 2

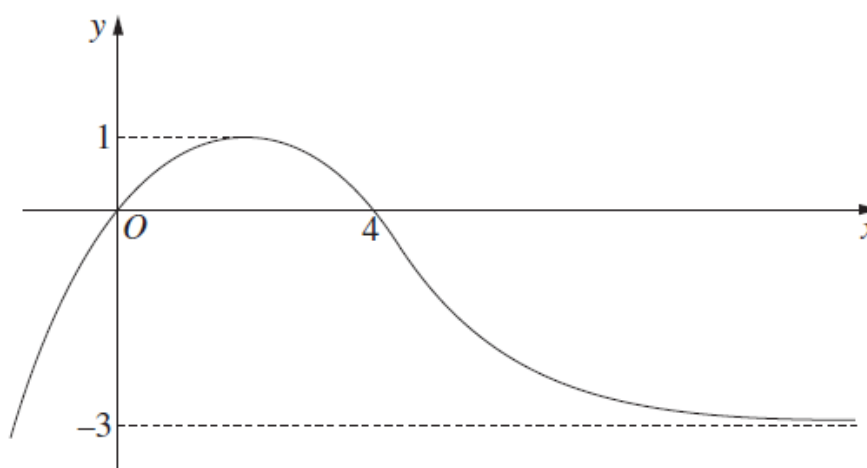
Question 4 (8 marks)

- (a) The sign of the function $y = h(x)$ changes either side of $x = a$. 2

Give two possible reasons as to what may happen at $x = a$ for this to occur.

- (b) Use the fact that $P(1) = 0$ to factorise $P(x) = x^4 + x^3 - 9x^2 + 11x - 4$ completely. 3

- (c) The diagram shows the graph of $y = f(x)$.

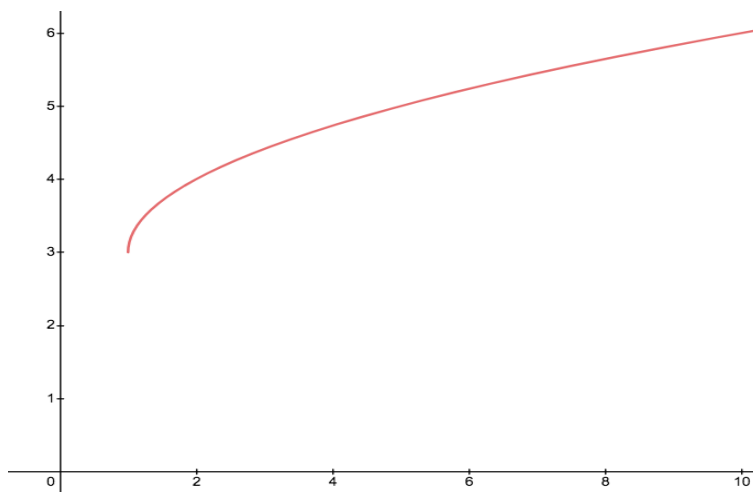


Draw separate one-third page sketches of the graphs of the following:

- (i) $y = [f(x)]^2$ 2
- (ii) $|y| = f(x)$ 1

Question 5 (8 marks)

- (a) The function $f(x) = 3 + \sqrt{x-1}$ is sketched below.



- (i) Explain why the inverse function $f^{-1}(x)$ exists. 1
 - (ii) Find a formula for $f^{-1}(x)$ noting any restrictions on the domain of $f^{-1}(x)$. 2
 - (iii) Hence or otherwise find the point(s) of intersection of $f(x)$ and $f^{-1}(x)$. 2
- (b) The cubic polynomial $P(x) = x^3 + rx^2 + sx + t$, where r , s and t are real numbers, has three real zeroes, 1, α and $-\alpha$.
- (i) Find the value of r . 1
 - (ii) Find the value of $s + t$. 2

Question 6 (10 marks)

- (a) Consider the function $f(x) = |5 - x| + |x - 2|$.
- (i) Find $f(0)$. 1
- (ii) For what values of x is $f(x)$ constant? 1
- (iii) What is the value of $f(x)$ for these values of x ? 1
- (iv) Draw a neat sketch of $y = f(x)$ showing any intercepts with the axes. 2
- (v) Hence or otherwise solve $|5 - x| + |x - 2| > 7$. 1
- (b) Consider the polynomial $P(x) = ax^4 + bx^3 + cx^2 + dx + e$ where a, b, c, d and e are all integers.
- (i) Suppose $\beta \neq 0$ is an integer such that $P(\beta) = 0$. 2
- Prove that $\frac{e}{\beta}$ is also an integer.
- (ii) Deduce that $Q(x) = 4x^4 - x^3 + 3x^2 + 2x - 3$ 2
does not have an integer zero.

End of Assessment Task

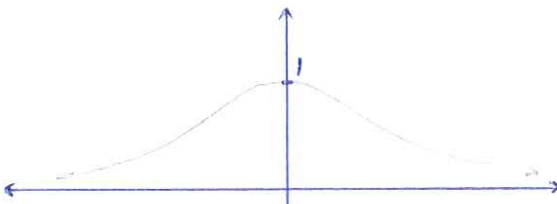


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Suggested Solution (s)	Comments
Question 1: (a) (i) $x^3 - 3x + \frac{3}{x} - \frac{1}{x^3}$	R/W
(ii) $x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)^3 + 3\left(x - \frac{1}{x}\right)$ $= 5^3 + 3(5)$ $= 140$	Aw 1 for $5^3 + 3 \times 5$
(b) one to many	R/W
(c) $x = y^2 - 6y$ $x + 9 = y^2 - 6y + 9$ $x + 9 = (y - 3)^2$ $y - 3 = \pm \sqrt{x + 9}$ $y = 3 \pm \sqrt{x + 9}$ Note: Accept: $y = \frac{6 \pm \sqrt{4x + 36}}{2}$ by use of the quadratic formula	Either expression for y Aw 1.
(d) (i) $f(g(x)) = (\sqrt{x})^2 - 3(\sqrt{x}) + 2$ $= x - 3\sqrt{x} + 2$	+ Aw 1
(ii) $(\sqrt{x})^2 - 3(\sqrt{x}) + 2 \geq 0$ $(\sqrt{x} - 2)(\sqrt{x} - 1) \geq 0$ $\sqrt{x} \leq 1, \sqrt{x} \geq 2$ But $x \geq 0 \implies x \geq 4$ $\therefore 0 \leq \sqrt{x} \leq 1$ $0 \leq x \leq 1$	Graphically: $x - 3\sqrt{x} + 2 \geq 0$ $\therefore x + 2 \geq 3\sqrt{x}$ $\therefore 0 \leq x \leq 1, x \geq 4$ 1 for factorisation 1 for $\sqrt{x} \leq 1, \sqrt{x} \geq 2$ 1 for answer



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Suggested Solution (s)	Comments
Question 2: (a) $x^3 - 2x^2 + a \equiv (x+2)Q(x) + 3$ If $x = -2$ $(-2)^3 - 2(-2)^2 + a \equiv (-2+2)Q(-2) + 3$ $-16 + a = 3$ $a = 19$	R/W
(b) $\frac{5}{x-4} \geq x$ $x \neq 4$ $5(x-4) \geq x(x-4)^2$ $5(x-4) - x(x-4)^2 \geq 0$ $(x-4)[5 - x(x-4)] \geq 0$ $(x-4)(5 - x^2 + 4x) \geq 0$ $(x-4)(x^2 - 4x - 5) \leq 0$ $(x-4)(x-5)(x+1) \leq 0$  $x \leq -1, 4 < x \leq 5$ Alternatively: $\frac{5}{x-4} \geq x$ $x \neq 4$ Solve $\frac{5}{x-4} = x$ $5 = x^2 - 4x$ $0 = x^2 - 4x - 5$ $0 = (x-5)(x+1)$ $\therefore x = -1, 5$  $x \leq -1, 4 < x \leq 5$	1 for excluding $x = 4$ from solution 1 for mult. by $(x-4)^2$ OR 1 for setting $\frac{5}{x-4} = x$ 1 for final solution
(c) 	1 for graph Range: $0 < y \leq 1$ 1 for range
(d) $x = 2 \tan \theta$ $y = 3 \sec \theta$ $\tan \theta = \frac{x}{2}$ $\sec \theta = \frac{y}{3}$ Since $1 + \tan^2 \theta = \sec^2 \theta$ $1 + \frac{x^2}{2^2} = \frac{y^2}{3^2}$ $\frac{y^2}{9} - \frac{x^2}{4} = 1$	1 for $\tan \theta$ & $\sec \theta$ 1 for answer



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Suggested Solution (s)	Comments
Question 3: (a) $2 \sin x \cos x = \sin x$ $2 \sin x \cos x - \sin x = 0$ $\sin x (2 \cos x - 1) = 0$ $\therefore \sin x = 0 \quad \text{or} \quad \cos x = \frac{1}{2}$ $x = 0^\circ, 180^\circ, 360^\circ \quad x = 60^\circ, 300^\circ$	1 for factorised equation 1 for each set of solutions $(0, 180, 360)$ $(60, 300)$
(b) (i) $\angle ACB = 55^\circ + 15^\circ$ or $\angle ACB = 180^\circ - 35^\circ - 75^\circ$ $= 70^\circ$ $= 70^\circ$	
(ii) In $\triangle ACD$, $\tan 15^\circ = \frac{h}{AC}$ $\therefore AC = \frac{h}{\tan 15^\circ}$ or $h \cot 15^\circ$ or $h \tan 75^\circ$ Similarly, in $\triangle BCD$, $BC = \frac{h}{\tan 17^\circ}$ or $h \cot 17^\circ$ or $h \tan 73^\circ$	1 for each expression
(iii) By cosine rule, $240^2 = h^2 \tan^2 75^\circ + h^2 \tan^2 73^\circ - 2(h \tan 75^\circ)(h \tan 73^\circ) \cos 70^\circ$ $= h^2 \tan^2 75^\circ + h^2 \tan^2 73^\circ - 2h^2 \tan 75^\circ \tan 73^\circ \cos 70^\circ$ $\therefore h^2 = \frac{240^2}{\tan^2 75^\circ + \tan^2 73^\circ - 2 \tan 75^\circ \tan 73^\circ \cos 70^\circ}$ $\therefore h \approx 59\text{m}$	1 for correct use of cosine rule 1 for 59m



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Question 4: (a) • A vertical asymptote exists at $x = a$ with $h(a-\epsilon)$ and $h(a+\epsilon)$ being opposite in sign • $h(x)$ is monotonic increasing or decreasing with $h(a) = 0$ (b) $P(x) = x^4 + x^3 - 9x^2 + 11x - 4$ $P(1) = 0 \therefore x-1$ is a factor $\therefore x^4 + x^3 - 9x^2 + 11x - 4 = (x-1)Q(x)$ By trial and error $P(-4) = 0$ $= (x-1)(x+4)R(x)$ $= (x-1)(x+4)(x^2 + kx + 1)$ $\therefore 4x - x - 4kx = 11x$ $-4kx = 8x$ $k = -2$ $\therefore P(x) = (x-1)(x+4)(x^2 - 2x + 1)$ $= (x-1)^3(x+4)$ Alternatively: $ \begin{array}{r} x^3 + 2x^2 - 7x + 4 \\ x-1 \overline{) x^4 + x^3 - 9x^2 + 11x - 4} \\ \underline{x^4 - x^3} \\ 2x^3 - 9x^2 \\ \underline{2x^3 - 2x^2} \\ -7x^2 + 11x \\ \underline{-7x^2 + 7x} \\ 4x - 4 \\ \underline{4x - 4} \\ 0 \end{array} $ $\therefore P(x) = (x-1)(x^3 + 2x^2 - 7x + 4)$ Then by trial and error & division again, $P(x) = (x-1)^2(x^2 + 3x - 4)$ or $P(x) = (x-1)(x+4)(x^2 - 2x + 1)$ $P(x) = (x-1)^3(x+4)$ $P(x) = (x-1)^3(x+4)$	← Award 1 ← Award 2 ← Award 3 ← 1 for cubic factor ← 2 for 2 linear factors 3 for answer

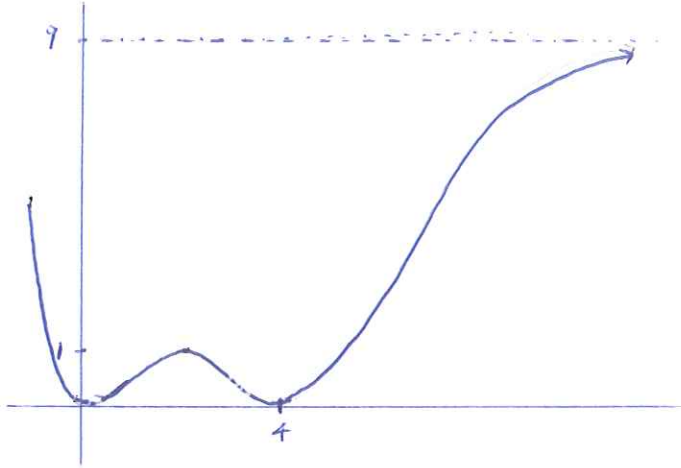


2019 - Year 11 Mathematics Extension 1 - Task 1 - SOLUTIONS

Suggested Solution (s)

Comments

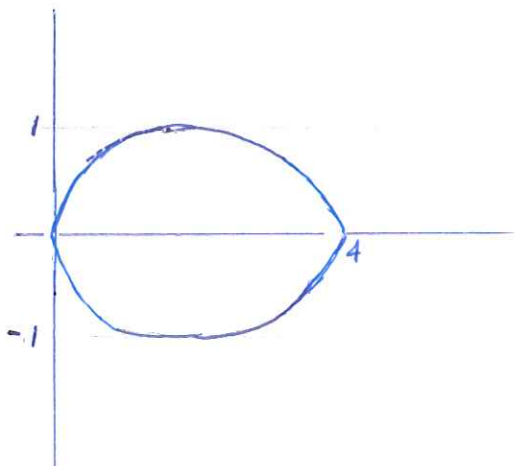
(c) (i)



1 for
correct
hor. asym.

1 for correct
turning pts

(ii)



1 for
graph
(must
be "pointy"
at (0,0)
& (4,0))



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Suggested Solution (s)

Comments

Question 5:

(a) (i) Because $f(x)$ is a one to one function or a monotonic increasing function

R/W

$$(ii) \quad x = 3 + \sqrt{y-1}$$

1 for
interchanging
 x & y

$$x - 3 = \sqrt{y-1}$$

$$(x-3)^2 = y-1$$

$$y = (x-3)^2 + 1 \quad (x \geq 3)$$

1 for
answer

$$(iii) \quad y = (x-3)^2 + 1$$

$$y = x$$

$$\therefore (x-3)^2 + 1 = x$$

$$x^2 - 6x + 10 = x$$

$$x^2 - 7x + 10 = 0$$

$$(x-2)(x-5) = 0$$

$$\therefore x = 2, 5$$

But $x \geq 3 \therefore$ discard $x = 2$

Point of intersection is $(5, 5)$

← 1 for
 x values

1 for $(5, 5)$

$$(b) \quad P(x) = x^3 + rx^2 + sx + t$$

$$(i) \quad \text{Sum of roots} = -r$$

$$\text{Sum of roots} = 1 + \alpha - \alpha = 1$$

R/W

$$\therefore r = -1$$

$$(ii) \quad P(1) = 0$$

1 for $P(1) = 0$

$$\therefore 1^3 - 1 + s \cdot 1 + t = 0$$

1 for
answer

$$\therefore s + t = 0$$



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Suggested Solution (s)	Comments
Question 6:	
(a) (i) $f(0) = 7$	R/W
(ii) $2 \leq x \leq 5$ (Accept $2 < x \leq 5$, $2 \leq x < 5$, $2 < x < 5$)	R/W
(iii) 3	R/W
(iv)	1 for correct graph 1 for (0,7)
(v) $x < 0$, $x > 7$	R/W
(b) (i) $P(x) = ax^4 + bx^3 + cx^2 + dx + e$ $P(\beta) = 0$ $\therefore 0 = a\beta^4 + b\beta^3 + c\beta^2 + d\beta + e$ $-e = \beta(a\beta^3 + b\beta^2 + c\beta + d)$ $\frac{e}{\beta} = -(a\beta^3 + b\beta^2 + c\beta + d)$ $\therefore$ since a, b, c, d and β are all integers, the RHS is an integer and, as such, $\frac{e}{\beta}$ is an integer.	1 for $\frac{e}{\beta}$ 1 for conclusion
(ii) We require the integer $\beta \neq 0$ to divide -3. i.e. $\beta = \pm 1$ or $\beta = \pm 3$ only Since, $Q(1) = 5 \neq 0$ $Q(-1) = 3 \neq 0$ $Q(3) = 327 \neq 0$ $Q(-3) = 369 \neq 0$ $\therefore Q(x)$ has no integer zero.	1 for possible β 's 1 for establishing $\beta \notin \mathbb{Z}$